A Hierarchical Bayesian Model to Analyze Big-Six Evolution in Premier League

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1 Introduction

Association football (soccer) is arguably the most popular sport globally, characterized not only by its immense commercial value but also by its inherent unpredictability. Unlike sports such as basketball, where high scoring frequencies average out random noise, football matches are low-scoring events where a single moment of brilliance or error can determine the outcome. This stochastic nature makes statistical modelling of football scores both a challenging and intellectually stimulating endeavour [1]. Among all competitions, the English Premier League (EPL) stands out as a focal point for analytics due to its rich history, financial dominance, and the availability of high-quality data.

Early approaches to modelling football scores relied on simple Poisson regression, assuming goal independence between competing teams [1]. However, these classical methods often struggle with limited data samples (a typical season consists of only 38 matches per team) and fail to fully capture the uncertainty in parameter estimates. To address these limitations, Baio and Blangiardo [2] proposed a Bayesian hierarchical framework that introduces partial pooling to stabilise inference. This approach treats team abilities (attack and defence strengths) as latent variables drawn from a common population distribution, enabling more robust estimation and natural uncertainty quantification via Markov Chain Monte Carlo (MCMC) methods.

In this paper, we conduct a comprehensive statistical analysis of the Premier League from the 1993–1994 season to the 2023–2024 season. We adopt the hierarchical Bayesian formulation to infer latent team parameters and evaluate the league's evolving dynamics. Our contributions are threefold:

1. **Tactical Profiling:** We perform an in-depth case study of the 1993–1994 season, demonstrating how the model decouples attack and defence strengths to characterise team playing styles.

2. **Longitudinal Analysis of Home Advantage:** We provide empirical evidence of the non-linear evolution of home advantage, highlighting its peak around the turn of the millennium and its recent decline.

3. **Quantification of Stratification:** By tracking the model's in-sample predictive accuracy over 30 years, we offer a novel metric for competitive balance. The observed increase in predictability serves as statistical evidence for the increasing stratification of the league.

The remainder of this paper is organised as follows. Section 2.2 describes the dataset and preprocessing steps. Section 3 details the mathematical formulation of the hierarchical Bayesian model and the MCMC inference procedure. Section 4 presents the results of the 1993–1994 case study and the longitudinal analysis. Finally, Section 6 concludes the paper and discusses future research directions.

2 Related Works and Data Processing

2.1 Poisson Models for Football Scores

Modeling football scores using the Poisson distribution has a long-standing tradition in sports analytics. The seminal work [1] demonstrated the reasonableness of idea that goals scored by each team can be modeled as independent Poisson random variables, with scoring intensities governed by their offensive and defensive strengths. Owing to its simplicity and interpretability, the model has remained a foundation in football analytics.

Subsequent enhancements improved realism and predictive accuracy. [3] incorporated a time-decay factor to emphasize recent matches and introduced a correction for low-scoring draws. Further refinements, such as [4], adjusted for seasonal dynamics including summer and winter breaks. Relaxing the assumption of independence between scores, [5] proposed bivariate Poisson models that can account for goal correlation, though univariate formulations remain common for their computational efficiency and empirical adequacy.

When modeling across multiple seasons, challenges such as team promotion and relegation arise. Surprisingly, not so many works deal with this problem explicitly. One common difficulty is the lack of historical data for newly promoted teams, making it hard to estimate their offensive and defensive strengths

2.2 Data Collection and Processing

We used the dataset [6], which contains detailed information on Premier League matches from 1993 to 2024. The dataset includes key match statistics such as date, home team, away team, final score, and other relevant variables.

To facilitate seasonal analysis and improve data handling, the original dataset was split into multiple CSV files, each corresponding to a single season. This allows for more structured processing and analysis across different time frames.

Basic data cleaning was also performed to ensure consistency, including removal of incomplete records and correction of minor formatting issues.

3 Modeling

3.1 Model Motivation

Football scorelines are naturally modeled using discrete count distributions, and the Poisson specification has been widely adopted in the statistical modelling of soccer results [2]. A Bayesian framework provides several advantages in this setting: (*i*) the number of matches per team is relatively small, so partial pooling through hierarchical priors stabilises inference; (*ii*) team abilities (attack and defence strengths) are latent quantities that can be regularised using weakly informative priors; (*iii*) posterior predictive simulation enables full uncertainty quantification for future match outcomes and league standings.

Following [2], we assume that each team has an underlying attack strength and defence strength, and that the home team enjoys a constant advantage which does not vary across teams or throughout the season.

3.2 Hierarchical Bayesian Poisson Model for Football Scores

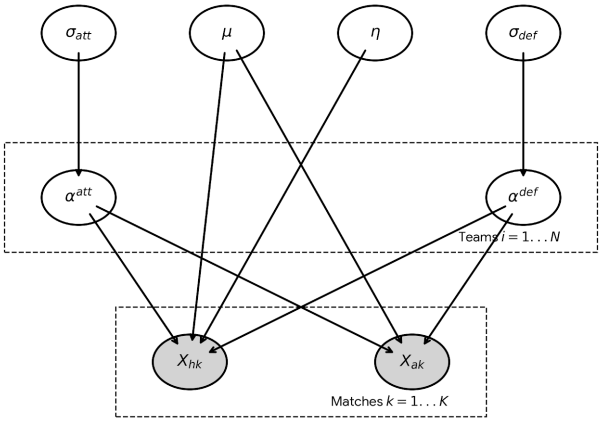


Figure 1: Plate Notation (DAG) of my model

3.2.1 Notation and Data Structure

Let N denote the number of teams and K the number of matches. For each match $k \in \{1, \ldots, K\}$, we observe the home goals X_{hk} and away goals X_{ak}. The indices $h_k, a_k \in \{1, \ldots, N\}$ denote the home and away teams, respectively.

We assume that the scoring intensity is determined by a global baseline, a home advantage term, and team-specific abilities. The parameters are defined as follows:

- μ: global intercept (baseline log-scoring rate),
- η: home advantage parameter (constant across teams),
- α_i^{att}: attack strength of team i (deviation from average),
- α_i^{def}: defence strength of team i (deviation from average).

3.2.2 Poisson Likelihood

Conditional on the parameters, the goals scored in the k-th match are assumed to follow independent Poisson distributions:

$$X_{hk} \mid \lambda_{hk} \sim \text{Poisson}(\lambda_{hk}), \qquad X_{ak} \mid \lambda_{ak} \sim \text{Poisson}(\lambda_{ak}).$$

The log-scoring intensities are modelled as linear combinations of the latent parameters. Crucially, we include a global intercept μ to capture the average league scoring rate, ensuring that the team-specific parameters represent relative deviations:

$$\begin{aligned} \log \lambda_{hk} &= \mu + \eta + \alpha_{h_k}^{\text{att}} - \alpha_{a_k}^{\text{def}}, \\ \log \lambda_{ak} &= \mu + \alpha_{a_k}^{\text{att}} - \alpha_{h_k}^{\text{def}}. \end{aligned} \tag{1}$$

Here, a positive defence parameter α^{def} implies a stronger defence, reducing the opponent's scoring rate.

3.2.3 Hierarchical Priors and Identifiability

To stabilise inference, especially for teams with limited data, we employ a hierarchical structure. The team-specific attack and defence parameters are modelled as exchangeable draws from normal populations centered at zero:

$$\alpha_i^{\text{att}} \sim \mathcal{N}(0, \sigma_{\text{att}}^2), \qquad \alpha_i^{\text{def}} \sim \mathcal{N}(0, \sigma_{\text{def}}^2), \quad i = 1, \ldots, N.$$

Since the model includes a global intercept μ, the location of the attack and defence parameters is not identifiable without constraints. We impose the standard sum-to-zero constraints to ensure interpretability:

$$\sum_{i=1}^{N} \alpha_i^{\text{att}} = 0, \qquad \sum_{i=1}^{N} \alpha_i^{\text{def}} = 0.$$

With these constraints, $\alpha_i^{\text{att}} > 0$ indicates an attack stronger than the league average.

3.2.4 Hyperpriors

We assign weakly informative priors to the global parameters and hyperparameters to allow the data to drive the inference while regularising extreme values.

The global intercept and home advantage are assigned normal priors:

$$\mu \sim \mathcal{N}(0, 2.5^2), \qquad \eta \sim \mathcal{N}(0, 2.5^2).$$

These priors cover a plausible range of log-goal rates (e.g., ±5 on the log scale covers realistic goal averages).

For the scale parameters (variances of team abilities), we avoid the inverse-gamma priors often used in older literature, as they can bias variance estimates away from zero. Instead, we use Half-Normal priors, which are robust and recommended for hierarchical models:

$$\sigma_{\text{att}} \sim \text{Half-Normal}(\sigma = 1), \qquad \sigma_{\text{def}} \sim \text{Half-Normal}(\sigma = 1).$$

3.2.5 Full Model Specification

The joint posterior distribution is proportional to the product of the likelihood and priors:

$$\begin{aligned} p(\Theta \mid \text{data}) \propto & \left[\prod_{k=1}^{K} \text{Poisson}(X_{hk} \mid \lambda_{hk}) \cdot \text{Poisson}(X_{ak} \mid \lambda_{ak}) \right] \\ & \times \prod_{i=1}^{N} \mathcal{N}(\alpha_i^{\text{att}} \mid 0, \sigma_{\text{att}}^2) \cdot \prod_{i=1}^{N} \mathcal{N}(\alpha_i^{\text{def}} \mid 0, \sigma_{\text{def}}^2) \\ & \times \mathcal{N}(\mu \mid 0, 2.5^2) \cdot \mathcal{N}(\eta \mid 0, 2.5^2) \\ & \times \text{Half-Normal}(\sigma_{\text{att}} \mid 1) \cdot \text{Half-Normal}(\sigma_{\text{def}} \mid 1). \end{aligned}$$

subject to the sum-to-zero constraints on α^{att} and α^{def}.

3.2.6 Posterior Inference using Markov Chain Monte Carlo

Given the hierarchical Bayesian model described in Section 3.2, the posterior distribution of the full parameter set

$$\Theta = \left\{\alpha^{\text{att}}_{1:N},\ \alpha^{\text{def}}_{1:N},\ \mu,\ \eta,\ \sigma_{\text{att}},\ \sigma_{\text{def}}\right\}$$

cannot be computed analytically. Therefore, we rely on simulation-based methods to approximate the posterior.

We obtain posterior samples using Markov Chain Monte Carlo (MCMC). Specifically, we employ the No-U-Turn Sampler (NUTS), a rigorous Hamiltonian Monte Carlo (HMC) algorithm, as implemented in the probabilistic programming library `PyMC`. NUTS efficiently explores the high-dimensional parameter space associated with the latent team abilities.

We run four parallel chains for 4,000 iterations each, discarding the first 1,000 iterations as burn-in to ensure the chains have reached their stationary distribution. Convergence is assessed using standard diagnostics:

- the Gelman–Rubin statistic $\hat{R}$, where values close to 1.0 indicate convergence,
- the effective sample size (ESS), ensuring sufficient independent draws,
- visual inspection of traceplots to detect any lack of mixing.

Posterior summaries are reported as means and 95% highest density intervals (HDI). Note that the team-specific parameters α^{att} and α^{def} are centered at each step (or deterministically transformed) to strictly satisfy the sum-to-zero constraints: $\sum \alpha^{\text{att}} = 0$ and $\sum \alpha^{\text{def}} = 0$.

3.2.7 Posterior Predictive Distribution

The Bayesian framework naturally propagates parameter uncertainty into predictions. For a future (or replicated) match between a home team h and an away team a, the posterior predictive distribution of the scoreline is given by:

$$p(\tilde{X}_h, \tilde{X}_a \mid \text{data}) = \int p(\tilde{X}_h, \tilde{X}_a \mid \Theta)\ p(\Theta \mid \text{data})\ d\Theta,$$

where $p(\tilde{X}_h, \tilde{X}_a \mid \Theta)$ represents the Poisson likelihood.

We approximate this integral via Monte Carlo simulation using the M posterior samples collected. For each posterior draw $\Theta^{(m)}$ (where $m = 1, \ldots, M$), we compute the expected scoring intensities, incorporating the global intercept $\mu^{(m)}$ and home advantage $\eta^{(m)}$:

$$\begin{aligned}
\lambda_h^{(m)} &= \exp\left(\mu^{(m)} + \eta^{(m)} + \alpha_h^{\text{att},(m)} - \alpha_a^{\text{def},(m)}\right), \\
\lambda_a^{(m)} &= \exp\left(\mu^{(m)} + \alpha_a^{\text{att},(m)} - \alpha_h^{\text{def},(m)}\right).
\end{aligned} \tag{2}$$

Based on these intensities, we simulate the match outcome:

$$\tilde{X}_h^{(m)} \sim \text{Poisson}\left(\lambda_h^{(m)}\right), \qquad \tilde{X}_a^{(m)} \sim \text{Poisson}\left(\lambda_a^{(m)}\right).$$

The resulting collection of simulated scores allows us to estimate:

- outcome probabilities (Home Win, Draw, Away Win),
- the most likely exact scorelines (e.g., 1–0, 1–1),

- total goal distributions.

To validate the model, we perform Posterior Predictive Checks (PPC) by comparing the distribution of the simulated data $\tilde{X}$ against the observed data X. Consistency between the observed statistics (e.g., the frequency of draws or the average number of goals) and the posterior predictive intervals indicates an adequate model fit.

4 Results

4.1 Model Validation and Convergence

Posterior inference was performed using NUTS with 4 chains of 2,000 iterations each (after 1,000 burn-in steps). Convergence diagnostics confirm the reliability of the estimates:

- The Gelman-Rubin statistic $\hat{R}$ was 1.00 for all parameters, indicating perfect mixing of the chains.
- The effective sample size (ESS) was sufficient for stable estimation of the 95% Highest Density Intervals (HDI).
- Posterior Predictive Checks (PPC) demonstrated that the model-generated data replicates the observed distribution of goals, including the frequency of clean sheets and high-scoring games.

To ensure the reliability of our Bayesian inference, we meticulously examined the convergence of the Markov Chain Monte Carlo (MCMC) algorithm. Figure 2 displays the trace plots and kernel density estimates (KDE) for the four global hyperparameters: the baseline intercept (μ), the home advantage parameter (η), and the standard deviations for the team-specific attack (σ_{att}) and defence (σ_{def}) effects.

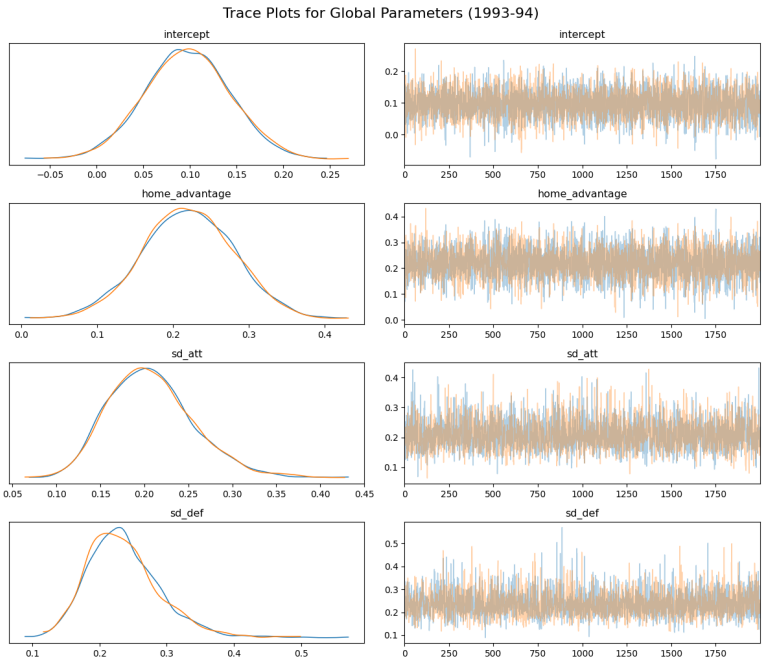


Figure 2: Trace Plot for Global Parameters(1993-1994)

We further inspected the convergence of latent team abilities by selecting three representative clubs from the 1993–1994 season: Manchester United (Champions), Arsenal (Defensive Specialists), and Swindon Town (Relegated). Figure 3 illustrates the trace plots for their respective attack (α^{att}) and defence (α^{def}) parameters.

The results highlight the distinct profiles of these teams. Critically, the trace plots for these high-dimensional latent variables remain stable and well-mixed, indicating that the hierarchical structure successfully pooled information without hindering the identifiability of individual team effects.

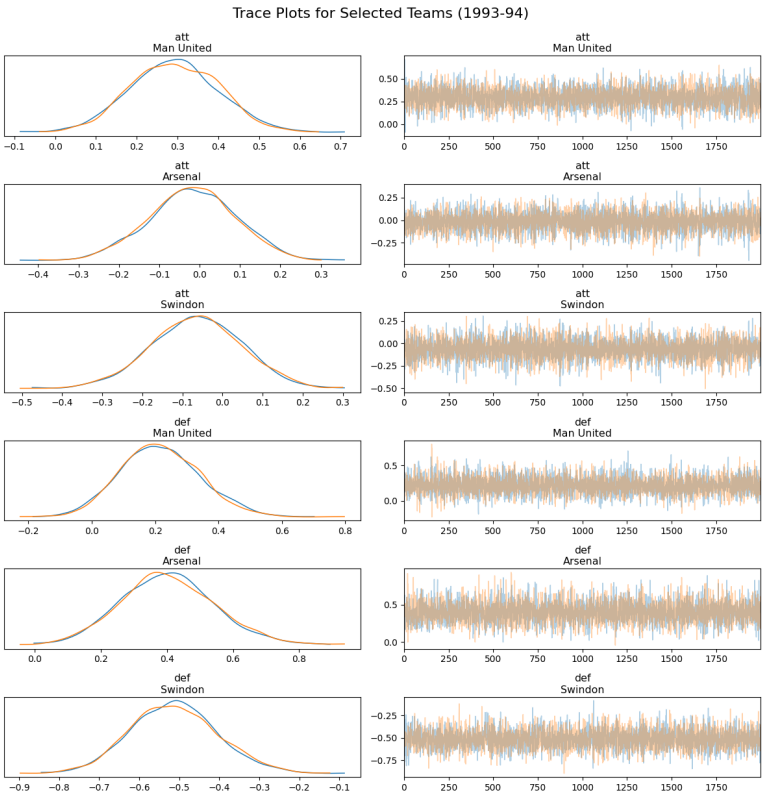


Figure 3: Trace Plot for Selected Teams

The PPC plots Figure 4 show that the model can generate synthetic match scores that look almost indistinguishable from real Premier League scores, meaning our Poisson–hierarchical structure fits the data very well.

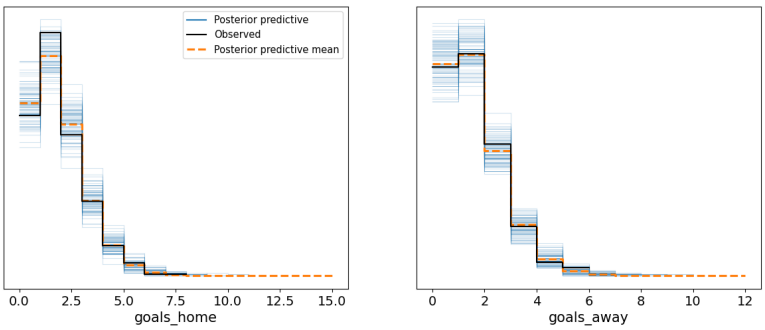


Figure 4: Posterior Predictive Checks for Home/Away Goals

4.2 Team Ability Estimation

Table 1 presents the estimated latent parameters for the top 5 and bottom 2 performing teams. The "Overall Strength" is calculated as the sum of the attack (α^{att}) and defence (α^{def}) parameters, representing the net log-goal advantage a team possesses against an average opponent.

Table 1: Estimated latent parameters for selected teams (1993-1994 Season). Teams are sorted by Overall Strength. Note: A positive Defence parameter indicates strong defensive capability (suppressing opponent scoring).

Team	**Attack** (α^{att})	**Defence** (α^{def})	**Overall**
Top Contenders			
Man United	0.295	0.220	**0.515**
Newcastle	**0.316**	0.170	0.485
Arsenal	-0.017	**0.403**	0.387
Blackburn	0.110	0.258	0.367
Leeds	0.132	0.209	0.342
Relegation Struggle			
Oldham	-0.151	-0.181	-0.332
Swindon	-0.066	-0.516	-0.582

4.2.1 Tactical Characterisation of Teams

The hierarchical model successfully decouples attack and defence capabilities, revealing distinct tactical profiles among the top teams (see Figure 6):

1. **The Balanced Champion (Manchester United):** Ranked first overall (Overall = 0.515), Manchester United exhibits a highly balanced profile. They possess the second-highest attack intensity and a top-tier defence, statistically validating their status as league champions.

2. **The "Entertainers" (Newcastle United):** The model identifies Newcastle as having the most potent attack in the league ($\alpha^{\text{att}} = 0.316$), surpassing even the champions. However, their defence ($\alpha^{\text{def}} = 0.170$) is weaker than that of their title rivals. This quantitative finding aligns with the historical narrative of Newcastle's aggressive, attacking style of play during the 1993/94 season.

3. **Defensive Specialists (Arsenal):** A striking insight is provided by Arsenal's parameter estimates. Despite having a below-average attack ($\alpha^{\text{att}} < 0$), they rank third overall due to a formidable defence ($\alpha^{\text{def}} = 0.403$), which is significantly higher than any other team. The model correctly learns that Arsenal's success relied on suppressing opponent scoring rather than high-volume goal production.

4. **Defensive Fragility (Swindon):** At the bottom of the hierarchy, Swindon displays a catastrophic defensive parameter ($\alpha^{\text{def}} = -0.516$). This magnitude suggests they were statistically distinct from other relegation candidates, prone to conceding significantly more goals than the league average.

4.3 Visualising Uncertainty

Figure 5 (Forest Plot) illustrates the 95% credible intervals for team attack strengths. While the point estimates provide a ranking, the overlapping intervals reveal tiers of ability. For instance, while Newcastle and Manchester United have the highest mean attack strengths, their intervals overlap, suggesting comparable offensive power. Conversely, there is a clear separation between the top tier and the bottom tier (e.g., Ipswich, Man City), indicating a high probability that the performance difference is not due to random chance.

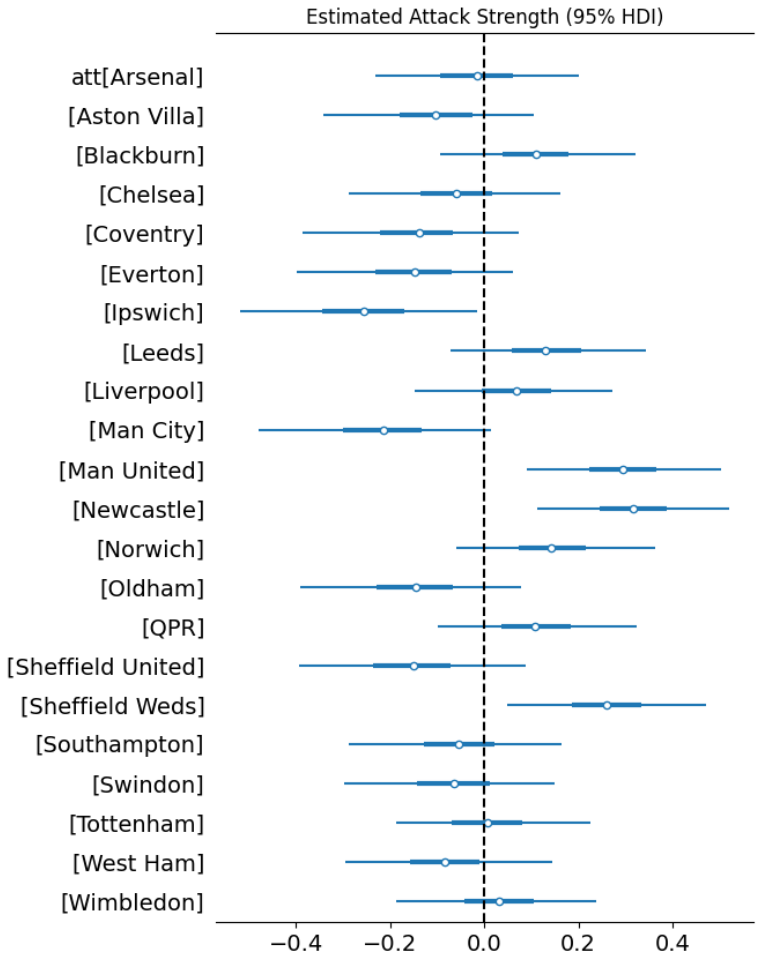


Figure 5: Forest plot of estimated Attack Strengths (α^{att}) with 95% Highest Density Intervals. Teams to the right are statistically stronger in attack.

Figure 6 plots Attack versus Defence strength. The separation of Arsenal (top-left quadrant: Low Attack, High Defence) and Newcastle (far right: High Attack) visually confirms the model's ability to discriminate between playing styles.

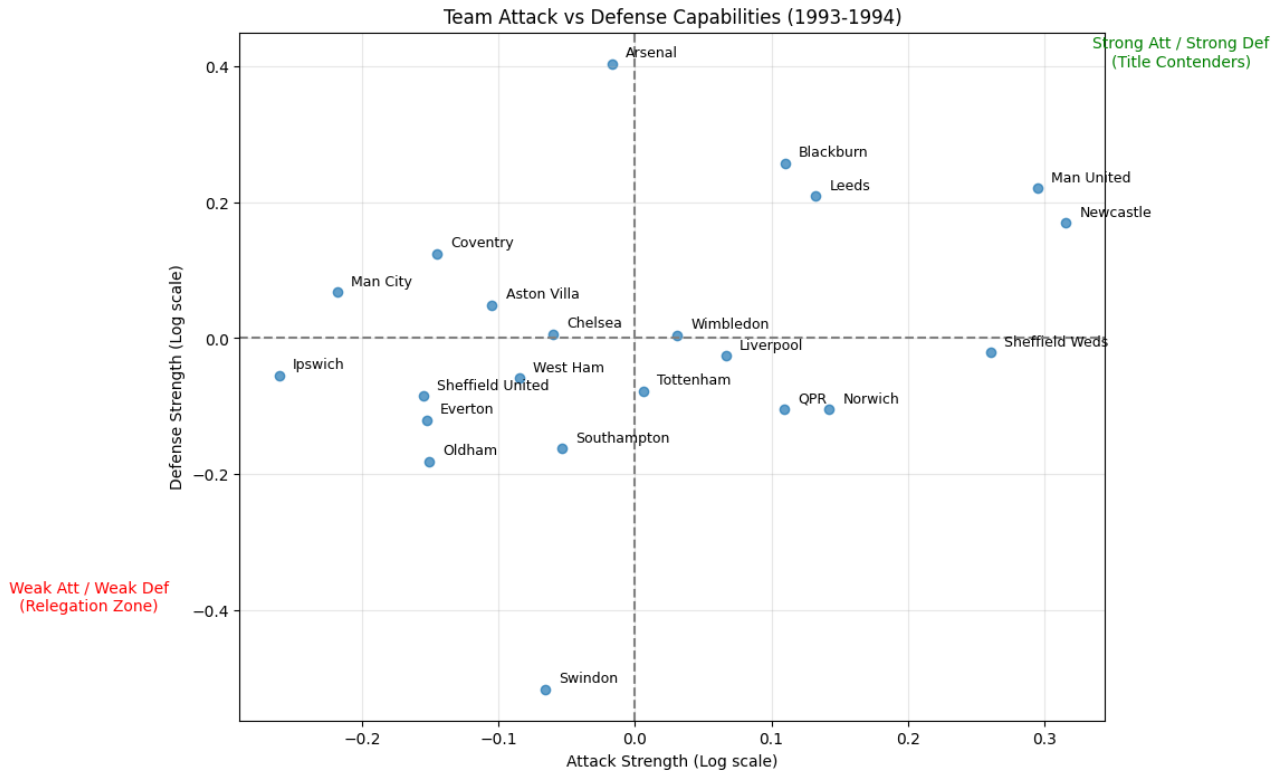


Figure 6: Bivariate plot of Attack vs. Defence parameters. The quadrants represent different tactical styles: Top-Right (Balanced Strong), Top-Left (Defensive Specialists), Bottom-Right (Attacking but Vulnerable).

5 Discussion: Longitudinal Analysis (1993–2024)

To investigate the structural evolution of the Premier League over the past three decades, we extended the Bayesian hierarchical framework calibrated in Section 3.2 to the entire dataset spanning from the 1993–1994 season to the 2023–2024 season. By fitting independent models to each of the 31 seasons, we tracked the temporal dynamics of two critical metrics: the predictability of match outcomes (Accuracy) and the magnitude of the home advantage (η).

The summary of these trends is visualised in Figure 7, and specific representative seasons are detailed in Table 2.

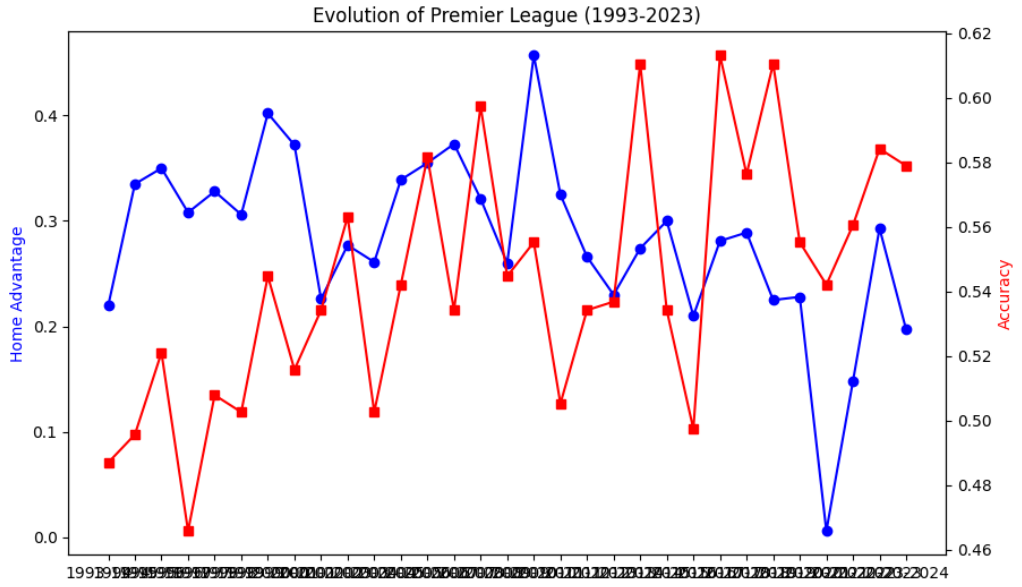


Figure 7: Longitudinal evolution of the Premier League (1993–2024). The red line (right axis) denotes the model's predictive accuracy, showing a general upward trend. The blue line (left axis) denotes the Home Advantage parameter η, highlighting a peak around 2000 and a subsequent decline.

5.1 The Crystallisation of Hierarchy: Analysis of Predictive Accuracy

The red curve in Figure 7 illustrates the *in-sample* predictive accuracy of the model, defined as the proportion of matches where the model's maximum a posteriori probability corresponds to the actual outcome ($y_k \in \{\text{Home}, \text{Draw}, \text{Away}\}$). Formally, for a total of K matches in a season, the accuracy $\mathcal{A}$ is given by:

$$\mathcal{A} = \frac{1}{K} \sum_{k=1}^{K} \mathbb{I}\left(y_k = \underset{c \in \{H,D,A\}}{\operatorname{argmax}} \hat{p}_{k,c} \right), \tag{3}$$

where $\hat{p}_{k,c}$ is the posterior predictive probability of outcome c for match k, and $\mathbb{I}(\cdot)$ is the indicator function.

Observation: We observe a clear upward trend in predictive accuracy, rising from a baseline of approximately 48.7% in 1993/94 to peaks exceeding 58% in the post-2020 era. While early seasons (e.g., 1996/97) exhibit significant volatility with accuracies dipping below 47%, the modern era shows a sustained high level of predictability.

Interpretation: This trend suggests a reduction in aleatory uncertainty within the league, likely driven by increasing financial and competitive stratification. In the early 1990s, the performance gap between teams was narrower, resulting in higher entropy in match results. In contrast, the modern Premier League is characterised by a dominant elite (often termed the "Big Six"). Structurally, this implies that the latent attack (α^{att}) and defence (α^{def}) parameters for top teams have diverged significantly from the league average, creating stronger signal-to-noise ratios for the Poisson likelihood, thereby making outcomes more deterministic.

5.2 The Erosion of the "Fortress": Dynamics of Home Advantage

The blue curve in Figure 7 tracks the evolution of the Home Advantage parameter η. In our log-linear formulation, η represents the additive increase in the log-scoring rate for the home team. The

Table 2: Longitudinal analysis of Home Advantage and Predictive Accuracy (1993–2024).

Season	**Home Adv (η)**	**Multiplier (e^{η})**	**Accuracy (%)**	**Context**
1993-1994	0.220	1.25	48.7%	League Inception (22 Teams)
1994-1995	0.335	1.40	49.6%	
1995-1996	0.350	1.42	52.1%	Reduction to 20 Teams
1996-1997	0.308	1.36	46.6%	
1997-1998	0.328	1.39	50.8%	
1998-1999	0.306	1.36	50.3%	
1999-2000	0.402	1.49	54.5%	Historical Peak Home Adv.
2000-2001	0.372	1.45	51.6%	
2001-2002	0.226	1.25	53.4%	
2002-2003	0.277	1.32	56.3%	
2003-2004	0.261	1.30	50.3%	Arsenal Invincibles
2004-2005	0.339	1.40	54.2%	
2005-2006	0.355	1.43	58.2%	
2006-2007	0.373	1.45	53.4%	
2007-2008	0.321	1.38	59.7%	Peak ‘Big Four’ Era
2008-2009	0.260	1.30	54.5%	
2009-2010	0.457	1.58	55.5%	Anomaly High HA
2010-2011	0.325	1.38	50.5%	
2011-2012	0.266	1.30	53.4%	Agüero Moment (High Drama)
2012-2013	0.230	1.26	53.7%	
2013-2014	0.274	1.32	61.1%	
2014-2015	0.300	1.35	53.4%	
2015-2016	0.210	1.23	49.7%	Leicester City Title (Outlier)
2016-2017	0.281	1.32	61.3%	
2017-2018	0.289	1.34	57.6%	Man City Centurions
2018-2019	0.225	1.25	61.1%	
2019-2020	0.228	1.26	55.5%	Season Interrupted
2020-2021	0.006	1.01	54.2%	COVID-19 (Empty Stadiums)
2021-2022	0.148	1.16	56.1%	Post-Pandemic Recovery
2022-2023	0.293	1.34	58.4%	High Stratification
2023-2024	0.197	1.22	57.9%	

multiplicative effect on the expected number of goals, denoted as $\mathcal{M}_{\text{home}}$, is derived as:

$$\lambda_{\text{home}} = \exp(\mu + \alpha^{\text{att}}_{\text{home}} - \alpha^{\text{def}}_{\text{away}} + \eta) = \lambda_{\text{neutral}} \times e^{\eta}. \tag{4}$$

Thus, e^{η} represents the multiplier applied to a team's scoring potential when playing at home.

Observation: Our analysis reveals distinct phases in the evolution of home advantage:

1. **Peak Era (1999–2005):** The home advantage reached its historical maximum around the turn of the millennium. Notably, in the 1999–2000 season, we estimated $\eta \approx 0.402$, which corresponds to a multiplicative boost of $e^{0.402} \approx 1.49$. In this era, home teams scored nearly 50% more goals than they would have on neutral ground, holding ability constant.

2. **Gradual Decline (2010–Present):** Since 2010, the parameter has shown a downward trajectory. In the most recent seasons, η has frequently dropped below 0.3, implying a significantly reduced advantage compared to previous decades.

Interpretation: The erosion of home advantage is consistent with broader findings in sports economics. Hypothesised drivers include the standardisation of pitch dimensions and quality, improved travel logistics reducing away team fatigue, and the introduction of Video Assistant Referee (VAR) technology, which may have mitigated subconscious referee bias towards the home crowd.

5.3 Robustness Check

It is important to note that the number of teams changed from 22 to 20 after the 1994-1995 season. The hierarchical priors employed in our model ($\alpha_i \sim \mathcal{N}(0, \sigma^2)$) effectively handled this structural change by pooling information, ensuring that the parameter estimates for Home Advantage and Accuracy remain comparable across different league formats.

6 Conclusion and Future Work

6.1 Conclusion

In this study, we implemented and extended the Bayesian hierarchical Poisson framework proposed by Baio and Blangiardo [2] to analyse the structure of football match outcomes. By applying this probabilistic model to the English Premier League, we derived insights at both the micro-level (team-specific tactical profiles) and the macro-level (longitudinal evolution of the league).

Our analysis of the 1993–1994 season demonstrated the model's ability to recover latent team characteristics from raw score data. The posterior estimates accurately identified the distinct playing styles of the era, such as the formidable defensive strength of Arsenal and the attack-heavy approach of Newcastle United. Furthermore, the hierarchical structure effectively handled the uncertainty inherent in sports data, providing robust credible intervals for team rankings.

Crucially, the longitudinal analysis spanning from 1993 to 2024 revealed significant structural shifts in the Premier League. First, we observed a steady increase in predictive accuracy (from $\approx 48\%$ to $\approx 59\%$), suggesting a "crystallisation of hierarchy" where the gap between elite and non-elite teams has widened, reducing aleatory uncertainty. Second, our results quantified the erosion of the home advantage effect, which peaked in the early 2000s ($\eta \approx 0.40$) but has declined significantly in the post-pandemic era. These findings highlight that football is not a static system; the underlying parameters governing match outcomes evolve dynamically over decades.

6.2 Future Work

While the independent Poisson model provides a robust baseline, several avenues exist to enhance predictive performance and model realism, particularly by integrating regression-based approaches and modern machine learning techniques.

6.2.1 Dynamic and Bivariate Extensions

The current model assumes that team abilities are constant within a season and that home and away goals are conditionally independent. Future research could incorporate time-varying parameters to account for form fluctuations and fatigue, similar to the dynamic bivariate Poisson models proposed by Koopman and Lit [4]. Additionally, explicitly modelling the correlation between home and away scores (e.g., a high-scoring home team might force the away team to open up) using a bivariate distribution could refine the likelihood function [5].

6.2.2 Hybrid Predictive Modelling (Regression and ML)

A primary goal for future work is to transition from explanatory modelling to pure forecasting of future seasons. We propose a hybrid approach where the latent parameters ($\alpha^{\text{att}}, \alpha^{\text{def}}$) estimated by our Bayesian model serve as high-quality feature inputs for regression or classification algorithms.

Recent literature suggests that combining statistical estimates with machine learning yields superior results. For instance, Groll et al. [7] demonstrated the efficacy of Random Forests in tournament prediction by incorporating estimated team abilities. Similarly, Baboota and Kaur [8] and Rodrigues and Pinto [9] highlight that Gradient Boosting and other ensemble methods can effectively capture non-linear interactions between team strength and external factors (e.g., injuries, transfers).

Therefore, our future work will focus on building a regression framework that utilizes the historical trajectory of our Bayesian parameters to forecast future league standings. By treating the posterior means of α^{att} and α^{def} as time-series data, we aim to predict the decline or rise of teams before the season commences, thereby bridging the gap between statistical inference and predictive analytics.

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